Machine Safety Specialist (CE)

Module 4: Fixed and interlocking guards

TAB4: Risk reduction and standards
Tab4-4/9 20240812

kader
Academy

Fixed and movable guards

EN-ISO 14120:2015

Nederlandse norm

NEN-EN-ISO 14120

(en)

Veiligheid van machines - Afschermingen - Algemene eisen voor het ontwerp en de constructie van vaste en beweegbare afschermingen (ISO 14120:2015,IDT)

Safety of machinery - Guards - General requirements for the design and construction of fixed and movable guards (ISO 14120:2015,IDT)

Guards

EN-ISO 14120

Machinery Regulation, Annex III, paragraph 1.4.1

- Guards and protective devices shall:
- Be firmly executed
- Be held securely in place
- do not pose any additional hazards
- cannot be easily circumvented
- are sufficiently far from the danger zone,
- obstructing the view of the work as little as possible
- Access options for work, if possible without dismantling the guard

Guards

EN-ISO 14120

Types of guards:

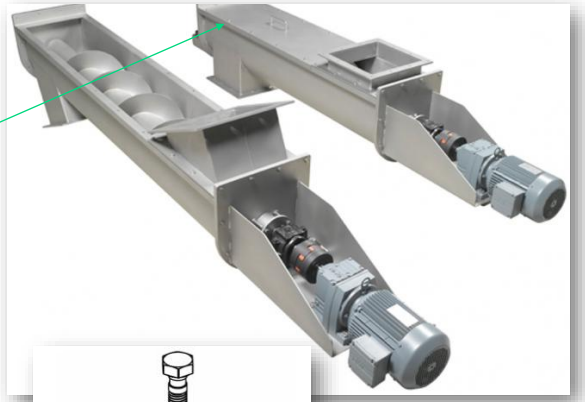
- Fixed guards
- Movable guard with guard locking
  (interlocking guard)
- Adjustable guard
- Interlocking guard with a start function

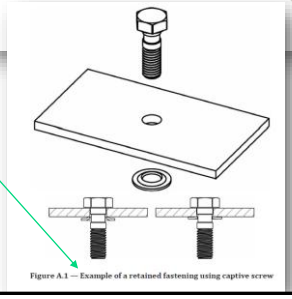
Tab4-4/9-Sh.5

Guards
EN-ISO 14120

Fixed guards

- Can only be removed with tools.
  (e.g. bolts)
- If possible, does not stay in place without the fasteners.
  (If all bolts are loose, the guard should not be able to stay in place if possible)
- Fasteners remain connected to the machine or the guard.
  (the bolts should not be completely removed to prevent loss for)





See also Machinery regulation, appendixIII, §1.4.2.1

Tab4-4/9-Sh.6

Guards
EN-ISO 14120

Movable guard with interlock
(Movable guards with locking device)

- Equipped with function-blocking device
  (For example: door switch, sensor, trapped-key)
- The operator must not be able to come near moving parts
  (Think of openings and safety distances)
- Locking and/or blocking
  (For example: detection door closed, then release for start machine)
- Still connected to the machine in the open position
  (hinges, or sliding door that can't be removed)

See also Machinery regulation, appendix III, §1.4.2.2

Guards
EN-ISO 14120

The choice of a fixed guard or movable guard with function lock for work by a maintenance engineer depends on the frequency of access (EN ISO 14120 6.4.4.1).

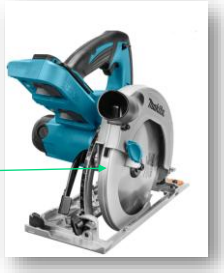
- Choose a movable guard with function lock at a high access frequency (for example: on average, access is required at least 1x per week)
- Fixed shielding can be justified with a low access frequency (for example: access is required on average less than 1x per week)

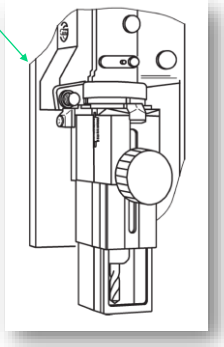
When an operator also needs access to a danger zone, fixed protection is not an option.

Guards
EN-ISO 14120

Adjustable guard

- Automatically adjustable (self-closing), or manually adjustable
- Tool-free adjustment
  (To be set by the operator himself)





See also Machinery regulation, appendixIII, §1.4.2.3

Guards

EN-ISO 14120

Interlocking guard with a start function(bedieningsscherm)

- Function-blocking guard where closing the guard leads to the start of the (dangerous) movement without using a separate start signal.

EN-ISO 12100 (chapter 6.3.3.2.5, interlocking guard with start function):
Requirements include:

- Meets at least all blocking screen requirements
- Cycle time short
- Unable to get trapped behind the shield
- All other guards (fixed and/or movable) executed as function-blocking guards.

Interlocking devices associated with guardsFeature-blocking guards

Nederlandse norm

NEN-EN-ISO 14119

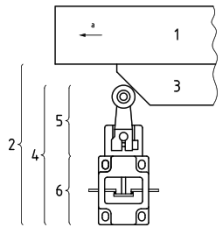
(en)

Safety of machinery - Interlocking devices associated with guards - Principles for design and selection (ISO 14119:2013,IDT)

EN ISO 14119

Interlocking device

- ‘switch" that blocks dangerous functions of a machine in certain situations (usually "when the gate is open")



Key

1	guard	4	position switch
2	interlocking device	5	actuating system
3	actuator	6	output system

[a] Direction of opening.

Figure 1 — Example of an interlocking device

Interlock; interlocking device

EN ISO 14119

Interlocking guard

Guard, coupled with an interlocking device that works with the control system to:

- Hazardous parts cannot function when the guard is open
- A stop command is given when the guard is opened
- The machine does not start when the guard is closed

Interlocking guard

EN ISO 14119

Interlocking guard with start function;
Start the dangerous machine functions without a separate start command

Additional requirements (EN ISO 12100)

- Short cycle times
- The guard may only be open for a short time (e.g. cycle time) after this time a new start command is required
- Small dimensions, no person should 'fit' behind the guard
- All other guards are function-blocking.
- Failure of the switch should not lead to an unexpected start-up
- The guard is reliably kept open, cannot close due to its own weight.

EN ISO 14119

Guard locking device

- Connected to the control system,
- Can hold a guard in the closed state.

Guard locking device

EN ISO 14119

Interlocking guard with guard-locking function

- Hazardous parts can only function when the guard is closed and locked.
- The guard remains closed and locked until the machine function no longer poses a risk

Interlocking guard with guard locking

Machinery Directive:

1.4.2.2. **Beweegbare afschermingen met blokkeervoorziening**

Wanneer het mogelijk is dat een bediener de gevarenzone bereikt voordat het risico dat voortvloeit uit de gevaarlijke machinefuncties is geweken, moeten de beweegbare afschermingen met blokkeervoorziening niet alleen met een vergrendelinrichting, maar ook met een voorziening voor het vergrendelen van de afscherming worden gecombineerd die:

Machinery regulation:

1.4.2.2. Beweegbare afschermingen met vergrendeling

Indien de bediener de gevarenzone kan bereiken voordat het risico dat uit de gevaarlijke functies van de machine of het verwante product voortvloeit, is geweken, moeten de beweegbare afschermingen niet alleen met een vergrendelinrichting, maar ook met een voorziening voor het vergrendelen van de afscherming worden gecombineerd die:

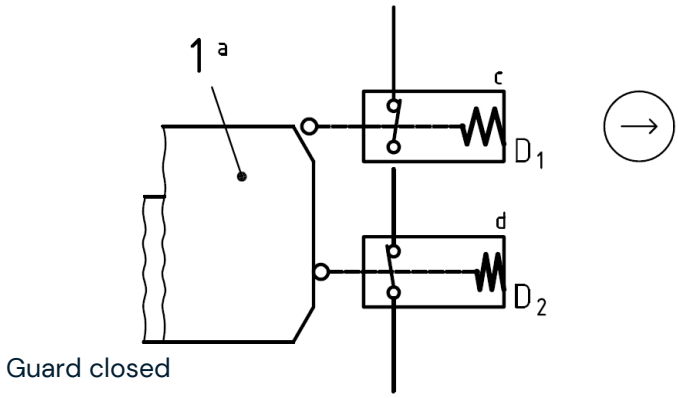
Where it is possible for an operator to reach the danger zone before the risk due to the hazardous machinery or related product functions has ceased, movable guards shall be associated with a guard locking device in addition to an interlocking device that:

Important principle

Direct mechanical action (3.10)
Positive mechanical action, movement of a mechanical component which arises inevitably from the movement of another mechanical component either by direct contact or via rigid elements

Direct opening action (3.11)
Positive opening operation. contact element achievement of contact separation as a direct result of a specified movement of the switch actuator through non-resilient members (for example not dependent upon springs)

Direct mechanical action & Direct opening action



Interlocking device types

Type 1	Mechanical	Uncoded	
Type 2	Mechanical	Coded	
Type 3	Non-contact actuated	Uncoded	
Type 4	Non-contact actuated	Coded	

Type 1, rotation

+

- Direct mechanical action
- Direct opening action
- Cannot be operated or manipulated manually

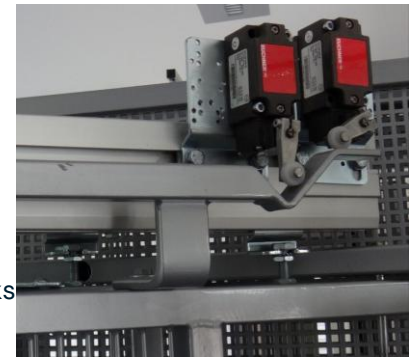
–

- Fails if the mechanical connection wears out or breaks
- Fails if alignment is not right

Type 1, linear

+

- Direct mechanical action
- Direct opening action
- Cannot be operated or manipulated manually



–

- Fails if the mechanical connection wears out or breaks
- Fails if alignment is not right

Type 1, Hinge

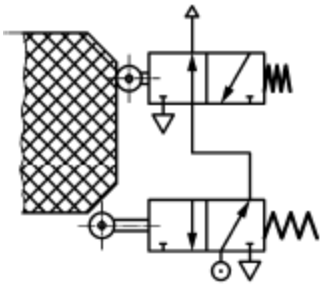
+

- Direct mechanical action
- Direct opening action
- Cannot be operated or manipulated manually
- 2 components in 1 (hinge+switch)

–

- Difficult to adjust

Type 1, pneumatic



Suitable for extreme conditions (EMC)

In combination with electrical switches (hybrid) resistant to common cause errors

Type 2, Switch with fork

+

- Direct mechanical action
- Direct opening action
- Easy to combine with a lock

–

- <u>Easy to bridge with spare fork</u>
- Damage due to misalignment
- Damage due to collision
- Sensitive to contamination (6.3.2)

Measures to prevent bridging necessary

Tab4-4/9-Sh.25

Type 2, trapped key

+

- No wiring to the shield
- Suitable for harsh environments
- Suitable for guards that can be completely removed
- Quick access is impossible (machine delay time))

–

- Can be bridged with a spare key
- Quick access is impossible

Coded keys required if there are multiple machines.

Tab4-4/9-Sh.26

Type 3, Contactless

Inductive, capacitive, magnetic, ultrasonic, optical. Not coded

+

- Contactless
- No moving parts
- Not sensitive to dust or liquid
- Easy to clean

–

- Uncoded
- Easy to bridge

Measures to prevent bridging necessary

Type 4, coded magnetic

+

- Compact
- Contactless
- Not sensitive to dust or liquid
- Easy to clean

–

- Low coding rate
- Bridging with spare part
- Often 'false alarm' in case of alignment problems
- Reed switches sensitive to shock and EMC

Type 4, coded RFID

+

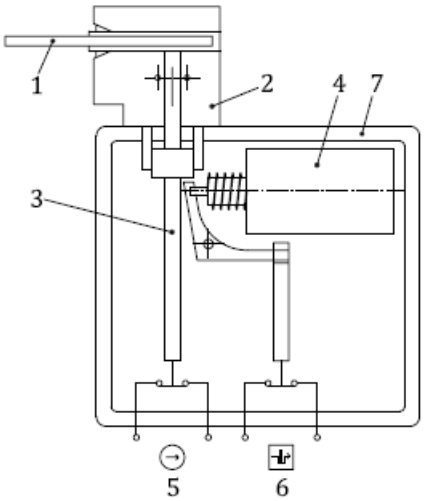
- Compact
- Contactless
- Not sensitive to dust or liquid
- Easy to clean
- Better coding
- No alignment problems

–

- Potentially sensitive to EMC

Locking

Spring applied – Power ON released



Locking

Always use

Spring applied – Power ON released
of
Power ON applied – Power ON released

Unless the risk assessment shows that this is not appropriate.

Power ON applied – spring released
Should only be used with equivalent level of safety.

Investigate power failure in particular.

Magnetic locking

- Monitor Holding Force (5.7.3.2)
- Monitor unauthorized opening(5.7.3.3)
  - with the response time penalty or going through a test cycle (>10 minutes)
- Holding force (5.7.3.4)
  - Attachment I
  - Test Requirement for Interlock Manufacturer

Magnetic locking

- Schmersal USA MZM100

Unauthorized Opening

ISO 14119 has adapted the monitoring functions of the MZM100 into the safety requirements for magnetic locking devices. Unlike conventional keyed locking devices, a magnet actuator can be forcefully detached from the switch under a locked state without damage to the unit. In order to deter against such manipulation of the device, the MZM100 with guard lock monitoring will go into a set 10 minute reset procedure if it detects an unauthorized/violent separation during a locked state.

Magnetic locking

- Personal security is often excluded by the supplier

www.euchner.de › en-us › products ▾ Vertaal deze pagina

Modular magnetic guard locking CEM-C60 | EUCHNER ...

Modular **magnetic guard locking** CEM-C60: Benefits for you650 N electromagnetic guard locking force for process protection, Numerous combination options ...

www.sick.com › safety-switches › s... ▾ Vertaal deze pagina

Safety locking devices | SICK

Safety switches with **locking** function **lock** protective devices securely and prevent access to ... Safety switch with **magnetic locking** device for process protection.

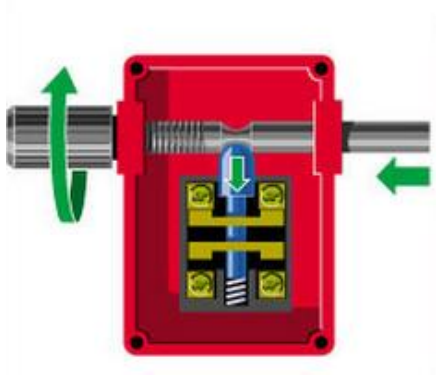
www.pilz.com › en-INT › eshop ▾ Vertaal deze pagina

Safety gate system PSENslock - Pilz INT

PSENslock with safety outputs that are independent of guard locking (Versions ... CL 3 of EN/IEC 62061 with **magnetic guard locking** for process protection tasks ...

2D drawing · 3D model · Certifications

Locking



Manual delay

Use and attachment of switches

a) Reliable attachment, can only be dismantled with tools.
b) Type 1 function lock must be removed after adjustment permanently fixed (e.g. penned).
c) Accessible for maintenance and adjustment, but taking manipulation into account.
d) Loosening must be prevented.
e) Manipulation must be prevented
f) Resistant to external influences due to position and possible protection.
g) Movement of mechanical switches must remain within the range defined by the manufacturer.
h) Not to be used as a stop/mechanical stop.
i) Clearance or switching moments must not lead to dangerous access through an opening (see also ISO 13855 and ISO 13857).
j) Support/attachment must be stable enough to allow correct use.

Use and attachment of actuators

a) Reliable attachment, can only be dismantled with tools.
b) Loosening must be prevented.
c) Resistant to external influences due to position and possible protection.
d) Not to be used as a stop/mechanical stop.
e) Support/attachment must be stable enough to allow correct use.

Use and attachment of lugs

a) Reliable attachment, can only be dismantled with tools.
b) Must be fixed permanently after adjustment (so no slotted holes).
c) May not affect the service life of the switch.

Escape release

Only accessible from inside the guard

- Hand-operated, works directly on the locking mechanism
- Causes a stop command

(5.7.5.2)

Tab4-4/9-Sh.39

Emergency release

- Zonder aanvullende voorwaarden
- Handbediend, werkt direct op het vergrendelmechanisme
- Veroorzaakt een stop commando
- Duidelijk gemarkeerd als noodvoorziening
- Niet zomaar te resetten, door ‘tool’ , component vervanging of besturingssysteem

(5.7.5.3)

Tab4-4/9-Sh.40

Auxiliary release

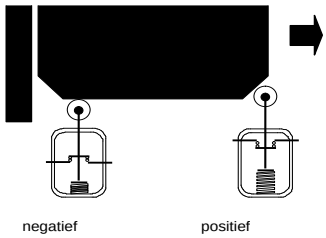
- Using a tool or key from outside the shield
- Only in exceptional cases such as power outages
- Causes a stop command
- Not easy to reset, by 'tool', component replacement or operating system

(5.7.5.4)

Operating the function lock

Operation of type 1 and type 2 switches:

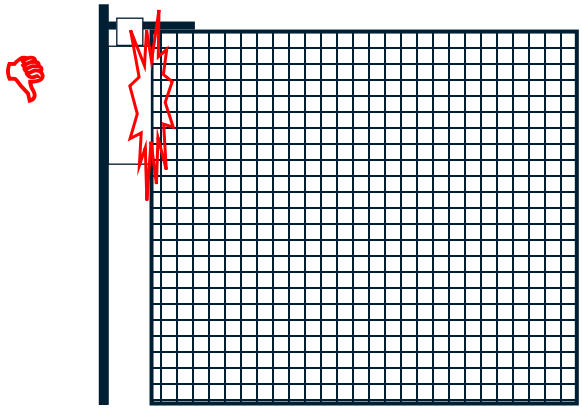
Mechanically forced, positive is mandatory; Not mechanically forced, negative is only allowed in combination with mechanically forced to avoid common cause errors.



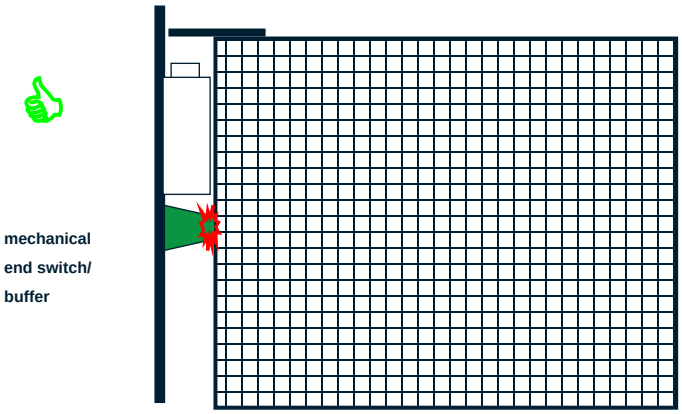
Choice of a lock

- Resistant to the expected forces
- Sturdy or powerful (electromagnet) enough to prevent the shield from opening. (without crowbar)
- Resistant to the dynamic effects
- If a door locks automatically, the lock must be able to absorb the resulting forces

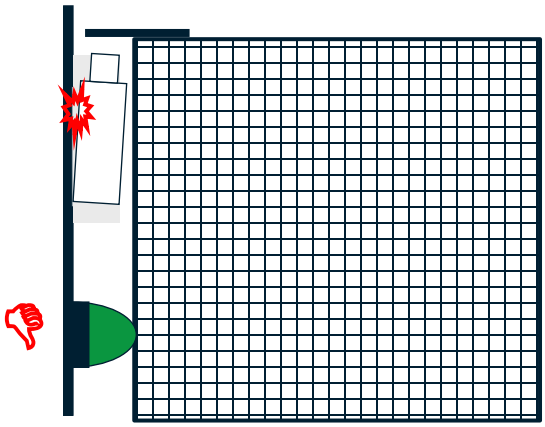
Switch not to be used as a stop



Dynamic load



Dynamic load



Manipulation

Some quotes

- According to the 940 respondents, almost 37% of the protective devices on machines are permanently or temporarily manipulated.
- In the opinion of all respondents, almost a quarter of all accidents at work are due to manipulation.
- It is also particularly tricky to estimate that manupulated machines or protective devices are tolerated in about a third of all companies.

Manipulation von Schutzeinrichtungen an Maschinen

HVBG
Hauptverband der gewerblichen Berufsgenossenschaften

http://www.dguv.de/ifa/de/pub/rep/pdf/rep05/manipulation_schutzeinrichtungen/ReportGesamt.pdf

Bypass

- Design the machine so that there is no reason to manipulate the function lock.
- Improving the reliability and operability (adjustment, malfunctions, etc.) of a machine is the most important step to prevent manipulation

Preventing manipulation

A

- Apply the aforementioned basic principles.
- Do not use a type 3 function lock unless the manipulation can be excluded (according to risk assessment)

Preventing manipulation

B

- Check that the operator has no reason to manipulate
  - through experience with previous systems
  - by the table in the annex H

Tab4-4/9-Sh.51

Evaluation of motivation to defeat

Task	Mode 1[a]	Mode 2[a]	Mode 3[a]	Mode 4[a]	Mode 5[a]	Task permissible in these modes of operation?	Task possible without defeating?	Easier/ more convenient[b]	Faster, increased productivity[b]	Flexibility, e.g. for larger workpieces[b]	Higher precision[b]	Better visibility[b]	Better audibility[b]	Less physical effort[b]	Reduced travel[b]	Greater freedom of movement[b]	Improved flow of movement[b]	Avoidance of interruption[b]	...
Initial operation																			
Program test/Test run																			
Setup/adjustment conversion/tooling/																			
Machining																			
Manual intervention for swarf removal																			
Manual change of workpiece																			
Manual intervention for trouble shooting																			
Checking/random sampling																			
Manual intervention for measuring/ fine tuning																			

[a] Modes of operation.

Op. Mode 1: .....

Op. Mode 2: .....

Op. Mode 3: .....

Op. Mode 4: .....

Op. Mode 5: .....

Modes of operation include automatic mode and manual mode, as in the example presented in Table H.2.

[b] Benefits without protective device: 0 = None; + = Minor; ++ = Substantial.

See Reference [15].

Tab4-4/9-Sh.52

Evaluation of motivation to defeat

Task	Mode 1[a]	Mode 2[a]	Mode 3[a]	Mode 4[a]	Mode 5[a]	Task permissible in these modes of operation?	Task possible without defeating?	Easier/ more convenient[b]	Faster, increased productivity[b]	Flexibility, e.g. for larger workpieces[b]	Higher precision[b]	Better visibility[b]	Better audibility[b]	Less physical effort[b]	Reduced travel[b]	Greater freedom of movement[b]	Improved flow of movement[b]	Avoidance of interruption[b]	...
Manual change of tools																			
Maintenance servicing																			
Rectification of faults machine																			
Cleaning, e.g. removal of swarf																			
...																			
...																			

[a] Modes of operation.

Op. Mode 1: .....

Op. Mode 2: .....

Op. Mode 3: .....

Op. Mode 4: .....

Op. Mode 5: .....

Modes of operation include automatic mode and manual mode, as in the example presented in Table H.2.

[b] Benefits without protective device: 0 = None; + = Minor; ++ = Substantial.

See Reference [15].

Preventing manipulation

C

- Take away the reason for manipulation for the operator as much as possible
  - Customize the design, add measures
  - Add operating modes (setup, tool change, etc.)

Preventing manipulation

D

- Take additional measures if the reasons for manipulation have not been completely eliminated.

Extra measures(7.2)

a) Reduce accessibility
->position, guarding, hiding

b) Make bridging technically difficult
->coded switches

c) Prevent switches from being removed
welding, 1-turn screws, rivets

Extra measures (7.2)

d) Prevent manipulation

1. 'Defeat monitoring'
   Status (check opening close in the machine cycle)
   Cyclic testing (system requires test-operation)
2. Plausibility check with additional (deviating) switch

Control systems

If high PL or SIL levels are required and function locks are rarely used, a manual function test should be carried out regularly

SIL3/HFT1 of PL e /cat3-4	Monthly
SIL2/HFT1 of PL d /cat3	Yearly

Automated function test is recommended

kader Academy